

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks: 50

Annexure A

APPLICATION-BASED QUESTIONS

Code of Conduct:

I SABEER KHAN M(192212101)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink, appearing to read 'M. Sabeer Khan' with a stylized flourish at the end.

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – ANSWER SCRIPT

Course Name: Cryptography and Network Security Course Code: CSA51

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security. (BL3, BL4)

Question 1: Event Communication Security (Caesar Cipher)

Scenario: A message “MEET AT NOON” must be encoded with a fixed shift of 4 positions (Caesar cipher) before being shared with core members.

(a) Reconstructing the encrypted message

The Caesar cipher encrypts a letter by shifting it forward in the alphabet by a fixed key k . Mathematically, for plaintext letter P :

$$C = (P + k) \bmod 26, \text{ where } A = 0, B = 1, \dots, Z = 25, \text{ and } k = 4$$

Applying a shift of 4 to each letter of “MEET AT NOON” (spaces are left unchanged):

Plaintext	M	E	E	T	A	T	N	O	O	N
Ciphertext	Q	I	I	X	E	X	R	S	S	R

Encrypted message: **QIIX EX RSSR**

(b) Retrieving the original message

An authorized member who knows the shift key ($k = 4$) reverses the process by shifting each ciphertext letter backward by 4 positions:

$$P = (C - k) \bmod 26 = (C - 4) \bmod 26$$

For example, Q (16) $- 4 =$ M (12); I (8) $- 4 =$ E (4); X (23) $- 4 =$ T (19), and so on. Applying this rule to every letter of “QIIX EX RSSR” restores the original text “MEET AT NOON”.

Because the Caesar cipher is symmetric, the same key is used for both encryption and decryption — only the direction of the shift changes. This is why the shift value must be shared secretly in advance (as a pre-agreed convention) among the core members, since anyone who knows k can trivially decrypt the message.

Question 2: Intercepted Suspicious Message (Monoalphabetic Substitution – Atbash Cipher)

Scenario: The intercepted ciphertext is “GSRH RH Z HVXIVG”. The word lengths (4, 2, 1, 6) and the repeated 2-letter word strongly suggest a simple substitution cipher rather than a modern block/stream cipher.

(a) Deducing the most likely original message

Testing a plain Caesar shift on the short 1-letter word “Z” fails to produce a meaningful common single-letter English word (‘A’ or ‘I’) for most shift values except a very specific mapping. Checking the mapping $G \rightarrow T$, $S \rightarrow H$,

$R \rightarrow I, H \rightarrow S$ shows that each ciphertext letter is the mirror image of the plaintext letter in the alphabet ($A \leftrightarrow Z, B \leftrightarrow Y, C \leftrightarrow X, \dots$), which is the rule of the Atbash cipher:

$$C = (25 - P) \bmod 26$$

Ciphertext	G	S	R	H	R	H	Z	H	V	X	I	V	G
Plaintext	T	H	I	S	I	S	A	S	E	C	R	E	T

Decoded message: THIS IS A SECRET

(b) Justification

- Word-length pattern: a 1-letter word decodes only sensibly to 'A' or 'I' in English — here $Z \rightarrow A$, which is consistent with a mirrored (reverse-alphabet) substitution.
- Repeated ciphertext word: "RH" appears twice (positions 2 and 5), decoding consistently both times to "IS" — a hallmark of a fixed monoalphabetic mapping where identical ciphertext always maps to identical plaintext.
- Structural check: applying the Atbash rule (first $\leftrightarrow$ last letter of alphabet) to every character produces grammatically correct English with no anomalies, confirming the cipher type.
- Because Atbash has no key (the mapping is fixed), it offers negligible security and is easily broken purely by pattern/frequency reasoning, without needing full frequency-count statistics.

Question 3: Secure Messaging System (Vigenère Cipher)

Scenario: The plaintext "HELLO" is encrypted using the repeating keyword "KEY".

(a) Step-by-step encryption

The Vigenère cipher combines each plaintext letter with a keyword letter using modular addition:

$$C = (P + K) \bmod 26, \quad A = 0, B = 1, \dots, Z = 25$$

The keyword "KEY" is repeated to match the length of "HELLO":

Position	1	2	3	4	5
Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
P value	7	4	11	11	14
K value	10	4	24	10	4
(P+K) mod 26	17	8	9	21	18
Ciphertext	R	I	J	V	S

Ciphertext: RIJVS

(b) Influence of the repeating keyword

Since the keyword “KEY” has only 3 letters but the message has 5, the keyword wraps around and repeats (K-E-Y-K-E). Each plaintext letter is shifted by a different amount depending on which keyword letter aligns with it (K = shift 10, E = shift 4, Y = shift 24). This means the same plaintext letter can encrypt to different ciphertext letters depending on its position — for example, the two L's in “HELLO” (positions 3 and 4) encrypt to J and V respectively because they align with different keyword letters (Y and K).

This position-dependent, polyalphabetic behaviour is what makes the Vigenère cipher considerably stronger than a simple Caesar or Atbash cipher: it flattens single-letter frequency patterns and resists basic frequency analysis, though it can still be broken using Kasiski examination or index-of-coincidence methods once the keyword length is guessed.

Question 4: Suspicious Digital Asset Investigation (Steganography)

Scenario: Investigators suspect a company logo image file hides confidential information that is not visible on casual inspection.

(a) Detecting hidden data within the image

- Statistical / histogram analysis: Comparing the image's colour-histogram and least-significant-bit (LSB) distribution against a known clean baseline; steganographic embedding tends to introduce abnormal statistical noise in the LSB plane.
- Chi-square and RS (Regular-Singular) steganalysis: Applying chi-square tests or RS analysis to detect deviations from expected LSB randomness that indicate hidden payloads.
- Visual/bit-plane inspection: Extracting and viewing individual bit planes (especially the LSB plane) using forensic tools, since hidden patterns sometimes become visible when isolated.
- File-property anomalies: Checking for unusual file size (larger than expected for the image's resolution/quality), unexpected metadata, or appended data after the normal end-of-file marker.
- Automated steganalysis tools: Running dedicated tools such as StegExpose, StegDetect, or Zsteg, which combine several of the above statistical tests to flag likely steganographic content.

(b) Two techniques to strengthen hidden communications against detection/extraction

1. Encrypt before embedding: Encrypt the secret payload (e.g., using AES) before hiding it in the image. Even if the hidden data is detected and extracted, it remains unreadable without the encryption key — combining cryptography with steganography (a layered defence).

2. Adaptive / randomized embedding: Instead of using sequential LSBs, use an adaptive embedding algorithm (e.g., F5 algorithm, matrix encoding, or a pseudo-random pixel-selection sequence driven by a secret key) that spreads the payload unpredictably across the image and preserves the original statistical profile, making chi-square/RS steganalysis far less effective.

Question 5: Legacy Encryption in Corporate Communication (Rail Fence / Transposition Cipher)

Scenario: A legacy transposition cipher with three levels of arrangement (a 3-row Rail Fence cipher) produced the ciphertext: “WECRLTEERDSOEFEAOCAIVDEN”.

(a) Reversing the encryption to find the original message

In a 3-rail Rail Fence cipher, the plaintext is written in a zig-zag pattern across 3 rows (down, then up, repeating), and the ciphertext is formed by reading the rows in order: Row 1 → Row 2 → Row 3.

The 25-letter ciphertext splits into rows as follows (row lengths follow the zig-zag pattern: 7, 12, 6):

Row	Letters read from ciphertext
Row 1 (7 letters)	W E C R L T E
Row 2 (12 letters)	E R D S O E E F E A O C
Row 3 (6 letters)	A I V D E N

Placing these letters back into their zig-zag positions (row pattern 0-1-2-1-0-1-2-1...) and reading row by row down the columns reconstructs the plaintext:

Original message: WE ARE DISCOVERED FLEE AT ONCE

(b) Why this method is vulnerable in modern systems

- Extremely small key space: the only “key” is the number of rails; for short-to-medium messages an attacker can simply try all reasonable rail counts (2, 3, 4, ...) and check which produces readable text — a trivial brute-force.
- No confusion, only diffusion of position: transposition alone does not disguise letter identity or frequency — the same letters and their frequencies remain, so once the correct arrangement is guessed, the plaintext is immediately recognisable English.
- Deterministic, unkeyed structure: without a secret keyword controlling column order, the zig-zag pattern is entirely predictable, so anyone who knows “it is a rail fence” can reconstruct the message purely by trial and error.
- Vulnerable to known-plaintext and pattern attacks: if any fragment of the plaintext (e.g., a common phrase) is guessed, the row structure can be solved immediately, exposing the rest of the message.

(c) Practical enhancement to improve confidentiality

Replace the simple fixed-rail transposition with a keyword-based columnar transposition (or combine transposition with substitution to form a product cipher, e.g., transposition followed by a Vigenère or AES encryption pass). A secret keyword determines both the number of columns and the order in which they are read out, dramatically increasing the key space and removing the fixed, guessable rail pattern. Applying multiple rounds of transposition (double transposition) or layering a modern symmetric cipher such as AES on top of the transposed text would bring the scheme up to a security level suitable for real corporate communication.